Immutable and Mutable Chameleon Showdown: A Comparative Study on openSUSE MicroOS and openSUSE Tumbleweed

Assessing system integrity, rollback, persistence resistance, and usability in transactional versus rolling-release openSUSE Linux

Firzan Irfandi Firman

Yogyakarta | October 3-4, 2026



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About Me

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Firzan Irfandi Firman (Irfan) is a cybersecurity professional specializing in security operations, threat detection, digital forensics, and enterprise security.

He currently works at Maybank in Security Operations and holds a Master of Information Technology in Cyber Security from the University of Technology Sydney, following a Computer Science bachelor's degree from Universitas Gadjah Mada.

His interests include cyber defense, digital forensics, cryptography, security operations and the evolving culture of cyberspace.

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Presentation Agenda

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01	*Introduction, Background and Motivation*
02	*Understanding The Main Concept*
03	*openSUSE MicroOS and openSUSE Tumbleweed in comparison*
04	*Conclusions drawn from this research*
05	*References*

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01 Introduction, Background and Motivation

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Immutable OS (as seen on the Internet)

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FROM LINUX » DISTRIBUTIONS

IncusOS: The new immutable and secure Linux distribution for Incus-based servers

Ver más

Instala Antivirus Profesional

Windows

Linux

Key points:

- IncusOS is an immutable distribution based on Debian 13 and desi Incus.
- It uses atomic A/B updates and full encryption with Secure Boot ar
- Administration without console or SSH, entirely via REST API.

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FUTURE OF WORK

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Tech > Services & Software > Operating Systems > Linux

One of the most user-friendly Linux distros I've ever used is also one of the most secure

Immutable Linux is the best path toward a truly secure operating system, and Aurora Linux might be the finest option for new users.

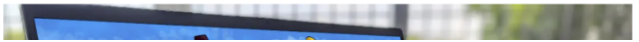


Jack Wallen
Contributing Writer

Jack Wallen is what happens when a Gen Xer mind-melds with present-day snark.... Read full bio

May 26, 2026, 2:27 pm ET

3 min read

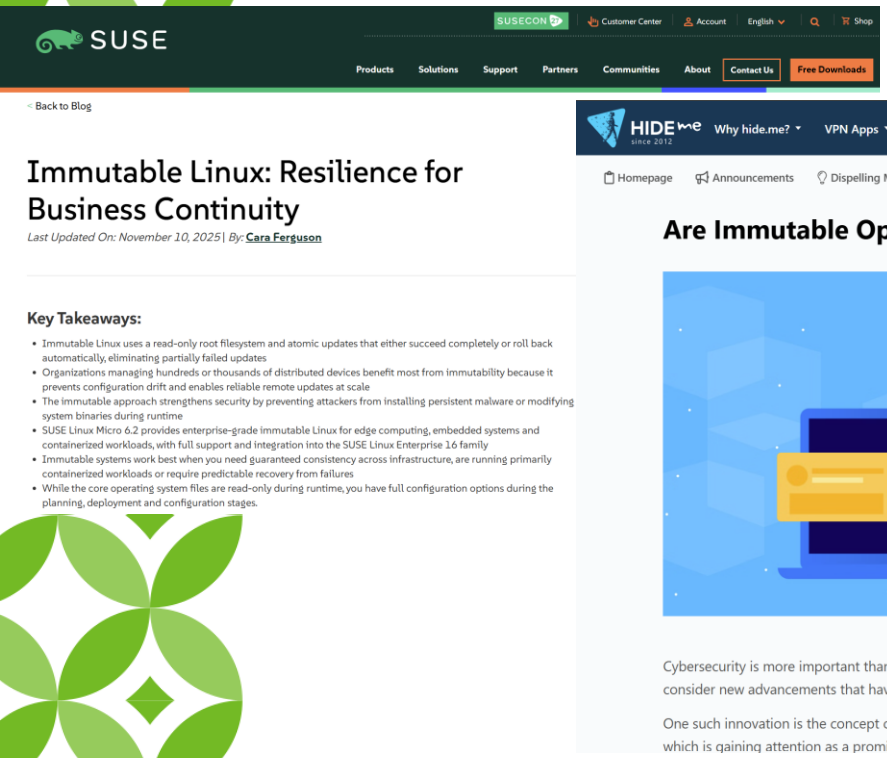


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Immutable OS (as seen on the Internet)

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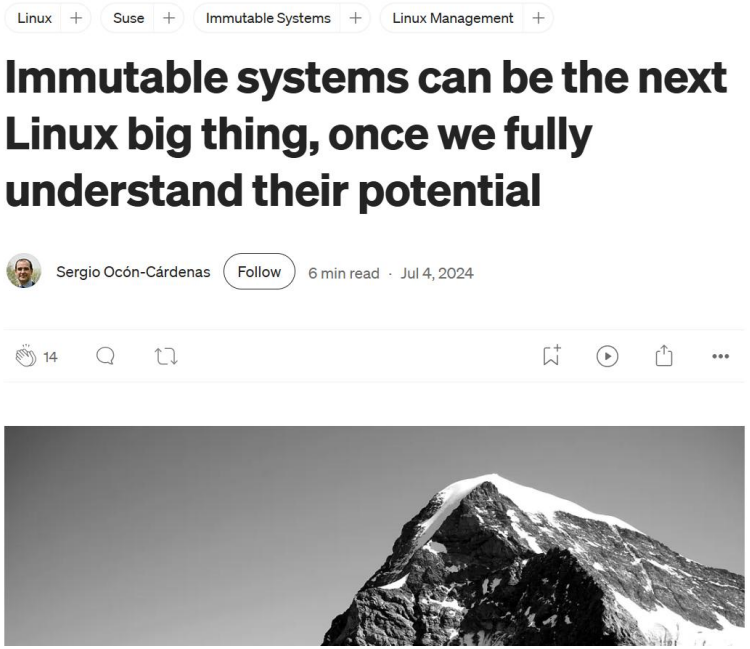


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Immutable OS (as seen on the Internet)

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Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.



Points found during my “scavenger hunt” on the beautiful World Wide Web …

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Security: Does OS immutability reduce unauthorized modification and malware persistence?

Configuration Drift: Does immutable Linux maintain more consistent system configurations than mutable Linux?

Update Reliability: Do transactional/atomic updates reduce failed or inconsistent system updates?

Recovery: Does built-in rollback enable faster recovery from failed updates or system corruption?

Reproducibility: Can immutable OS architectures produce more consistent and reproducible system states?

Performance & Resources: What are the CPU, memory, storage, and boot-time costs of immutability?

Flexibility & Compatibility: What limitations does immutability introduce for software installation, drivers, customization, and administration?

Overall Trade-off: Do the security and resilience benefits of immutable Linux outweigh its operational and compatibility costs?

Aha! The core research question !

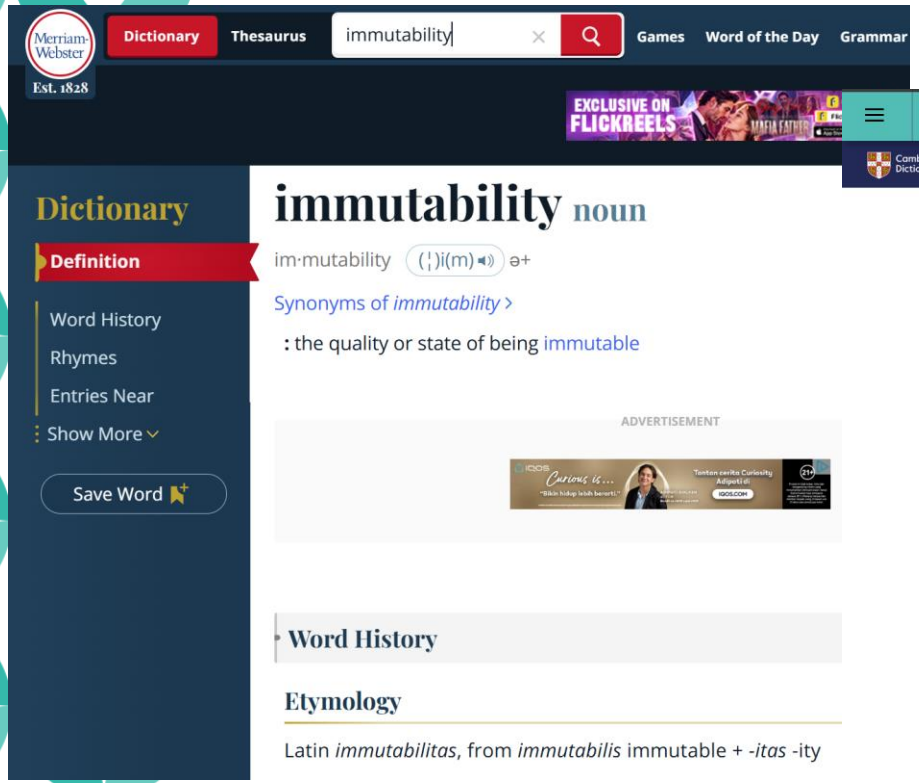
How do transactional and rolling-release openSUSE Linux systems differ in system integrity, rollback capability, persistence resistance, and usability?

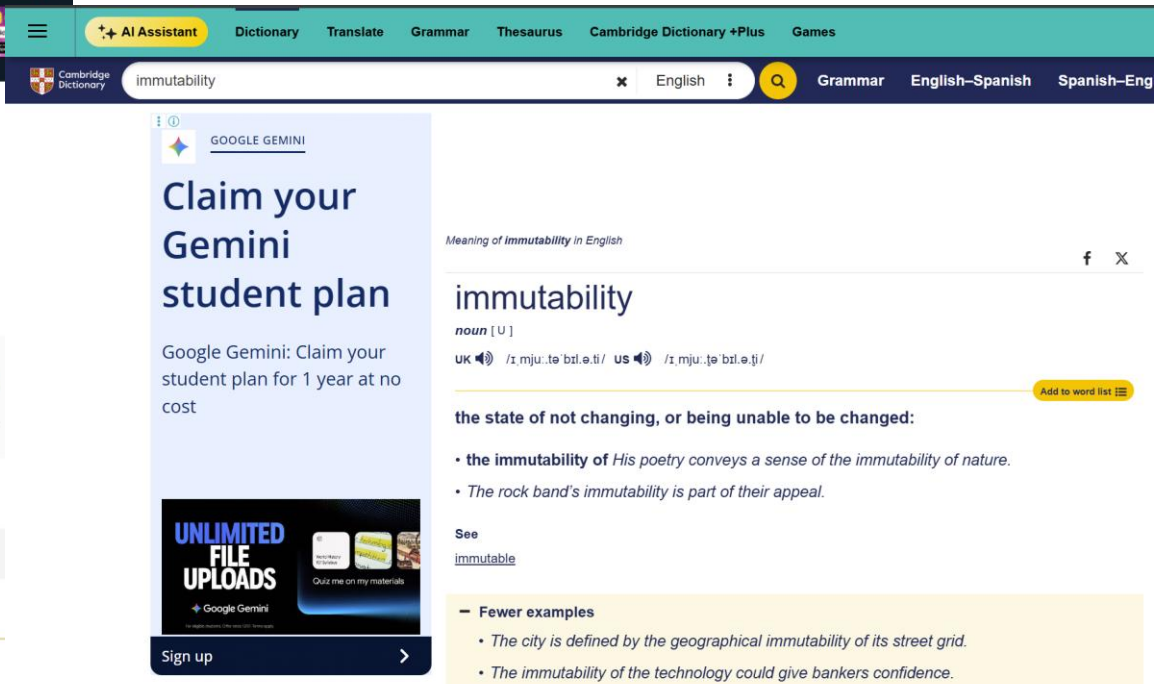
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02 Understanding The Main Concept

“Immutability” in terminology

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Busting the Jargon: What is an Immutable Operating System? (Part 1)

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An immutable operating system – particularly found on Linux distro – is an operating system with read-only at its core; in which system files and directories cannot be changed.



You can still add/remove personal files and do daily tasks normally; changes to data are stored as usual.



Any admin/system-level change either fails or is temporary and lost after a reboot

Busting the Jargon: What is an Immutable Operating System? (Part 2)

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Immutable OSes uses an "image-based" update method; the entire OS is replaced with a newer instance rather than individual packages being upgraded

It's not a unique OS: it's the same Linux, just with restricted modification and a different update model!

Busting the Jargon: What is an Immutable Operating System? (Part 3)

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Standard distros can be broken by a single sudo-level change and stay broken until fixed or reinstalled



Immutable distros can't be broken this way, offering more stability than even standard LTS releases



Updates guarantees your personal files intact while the update swapping out the system

Busting the Jargon: What is an Immutable Operating System? (Part 4)

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Attackers can't make permanent changes — a reboot restores system files, giving security by design

Updates are more reliable and safer, simplifying maintenance

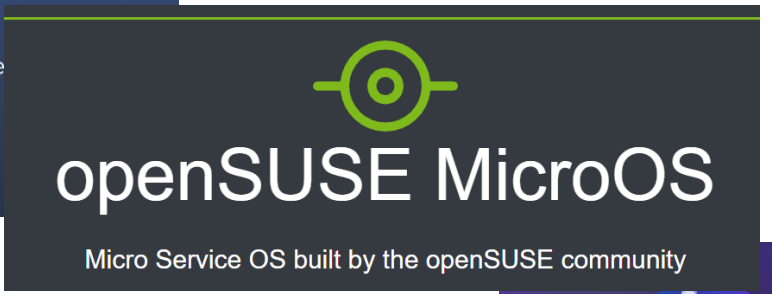
Prevents unintentional system changes (everything is read-only)

Current Immutable examples (as of now) in the GNU/Linux universe

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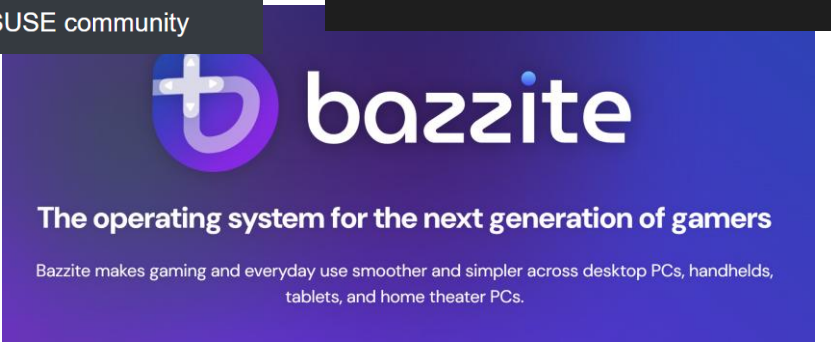












… and many more … !!!

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03 openSUSE MicroOS and openSUSE Tumbleweed in comparison

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Meet the Two Chameleons

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MicroOS

- Immutable
- Transactional
- Atomic Updates
- Rollback
- Controlled System State

Tumbleweed

- Mutable
- Rolling Release
- Direct Package Management
- Flexible Administration
- Continuous System Evolution



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So, what are we measuring exactly here?

- Root filesystem immutability
- Update mechanism
- Rollback capability
- Persistence resistance
- User-space resistance
- System drift
- Administrative usability

Meet the two chameleons...!

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MicroOS

New Tab Split View

```
microos@localhost:~> cat /etc/os-release
NAME="openSUSE MicroOS"
# VERSION="20260923"
ID="opensuse-microos"
ID_LIKE="suse opensuse opensuse-tumbleweed microos sl-micro"
VERSION_ID="20260923"
PRETTY_NAME="openSUSE MicroOS"
SYSEXT_LEVEL="glibc-2.44"
ANSI_COLOR="0;32"
CPE_NAME="cpe:/o:opensuse:microos:20260923"
BUG_REPORT_URL="https://bugzilla.opensuse.org"
SUPPORT_URL="https://bugs.opensuse.org"
HOME_URL="https://www.opensuse.org/"
DOCUMENTATION_URL="https://en.opensuse.org/Portal:MicroOS"
LOGO="distributor-logo-MicroOS"
microos@localhost:~>
```

Tumbleweed

Terminal - tumbleweed@localhost.localdomain:~

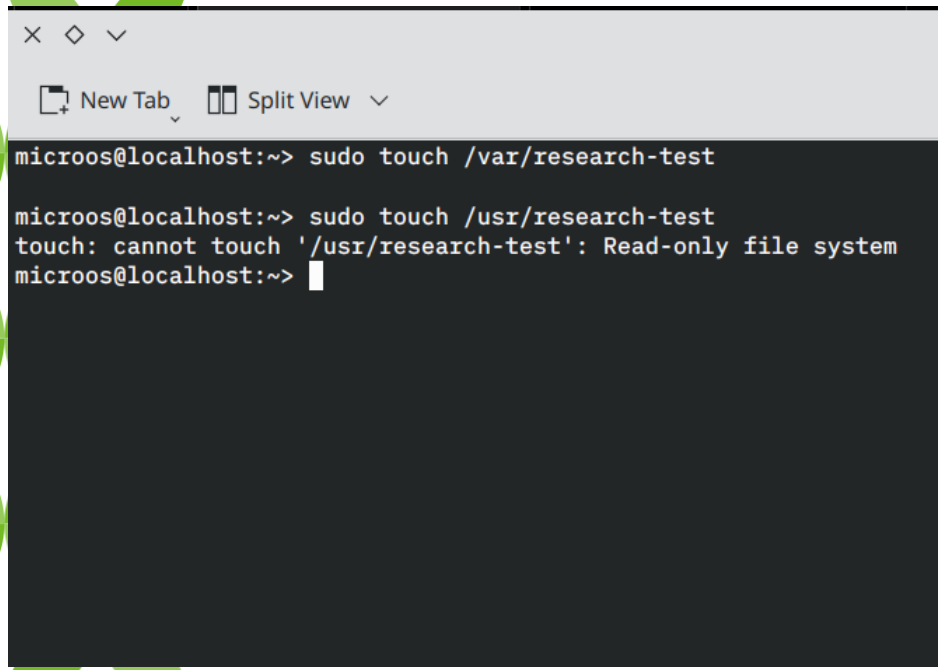
File Edit View Terminal Tabs Help

```
tumbleweed@localhost:~> cat /etc/os-release
NAME="openSUSE Tumbleweed"
# VERSION="20260919"
ID="opensuse-tumbleweed"
ID_LIKE="opensuse suse"
VERSION_ID="20260919"
PRETTY_NAME="openSUSE Tumbleweed"
ANSI_COLOR="0;32"
# CPE 2.3 format, boo#1217921
CPE_NAME="cpe:2.3:o:opensuse:tumbleweed:20260919:*:*:*:*:*:*:*"
#CPE 2.2 format
#CPE_NAME="cpe:/o:opensuse:tumbleweed:20260919"
BUG_REPORT_URL="https://bugzilla.opensuse.org"
SUPPORT_URL="https://bugs.opensuse.org"
HOME_URL="https://www.opensuse.org"
DOCUMENTATION_URL="https://en.opensuse.org/Portal:Tumbleweed"
LOGO="distributor-logo-Tumbleweed"
tumbleweed@localhost:~>
```

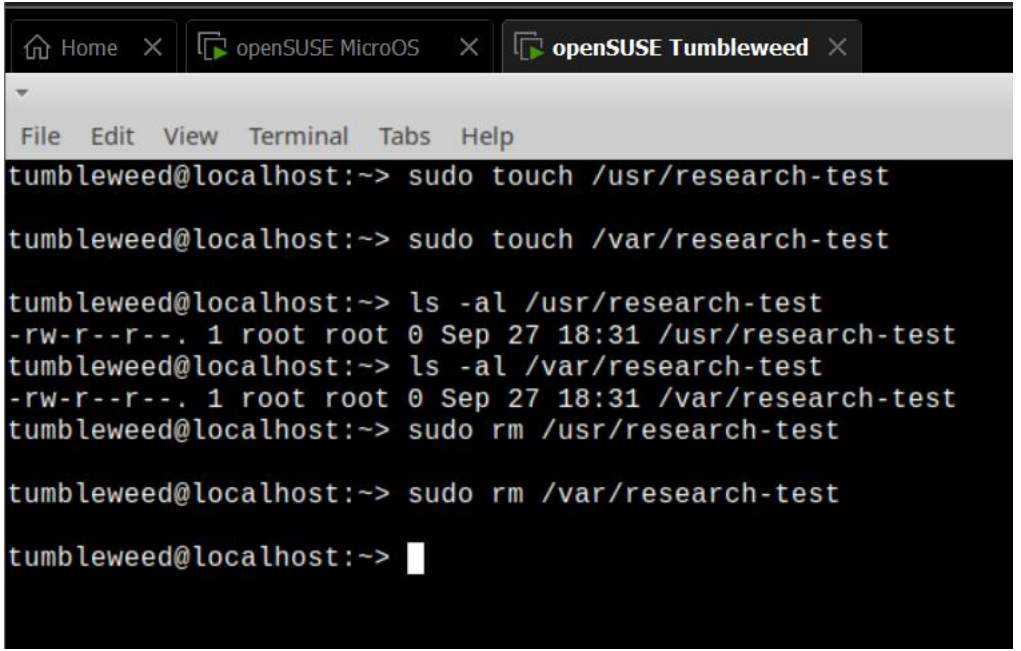
Root Filesystem Immutability

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MicroOS



Tumbleweed



Root Filesystem Immutability

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OMG! The system slaps your hand away!
You are root yet /usr refuses us!
But notice the subtlety in which /var, and others may still yield to our touch.

The point is:

- MicroOS does not trust even root with the core system. It treats the operating system image as sacred ground.
- Meanwhile, Tumbleweed treats the operating system as a moldable, shapeable, and alive in the hands of its administrator.
- MicroOS prioritizes system integrity; Tumbleweed prioritizes administrative freedom

Update Mechanism

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MicroOS

```
New Tab   Split View
microos@localhost:~> sudo snapper list
 #  | Type   | Pre # | Date                            | User | Used Space | Cleanup | Description           | Userdata
----+--------+-------+---------------------------------+------+------------+---------+-----------------------+---------
0   | single |       |                                 | root |            |         | current               |
1   | single |       | Mon 21 Sep 2026 08:16:42 AM UTC | root |   2.24 GiB |         | first root filesystem |
2   | single |       | Sun 27 Sep 2026 10:25:06 AM UTC | root |   2.63 GiB | number  | Snapshot Update of #1 |
3*  | single |       | Sun 27 Sep 2026 10:49:29 AM UTC | root |   2.64 GiB | number  | Snapshot Update of #1 |
microos@localhost:~> sudo transactional-update dup
tukit 6.1.3 started
Options: --log=console -c3 open
Using 'snapper' as snapshot manager.
Using snapshot 3 as base for new snapshot 5.
ID: 5
Transaction completed.
tukit 6.1.3 started
Options: --log=console abort 5
Using 'snapper' as snapshot manager.
Discarding snapshot 5.
Transaction completed.
microos@localhost:~> sudo transactional-update dup
tukit 6.1.3 started
Options: --log=console -c3 open
Using 'snapper' as snapshot manager.
Using snapshot 3 as base for new snapshot 6.
ID: 6
Transaction completed.
tukit 6.1.3 started
Options: --log=console abort 6
Using 'snapper' as snapshot manager.
Discarding snapshot 6.
Transaction completed.
microos@localhost:~>
```

Tumbleweed

```
Terminal - /usr/bin/zypper
File  Edit  View  Terminal  Tabs  Help
tumbleweed@localhost:~> sudo zypper refresh
Repository 'Main Update Repository' is up to date.
Repository 'Main Repository (NON-OSS)' is up to date.
Repository 'Main Repository (OSS)' is up to date.
Repository 'openSUSE-20260723-0' is up to date.
Repository 'Open H.264 Codec (openSUSE Tumbleweed)' is up to date.
All repositories have been refreshed.
tumbleweed@localhost:~> sudo snapper list
 #  | Type   | Pre # | Date                            | User | Used Space | Cleanup | Description
----+--------+-------+---------------------------------+------+------------+---------+-----------------------
0   | single |       |                                 | root |            |         | current
1*  | single |       | Mon 21 Sep 2026 09:02:55 AM WIB | root | 111.05 MiB |         | first root filesystem
2   | single |       | Mon 21 Sep 2026 09:41:40 AM WIB | root | 110.61 MiB | number  | after installation
3   | pre    |       | Sun 27 Sep 2026 05:24:35 PM WIB | root |   6.36 MiB | number  | zypp(zypper)
4   | post   |     3 | Sun 27 Sep 2026 05:24:41 PM WIB | root |   4.44 MiB | number  |
tumbleweed@localhost:~> sudo zypper dup
Loading repository data...
Reading installed packages...
Warning: You are about to do a distribution upgrade with all enabled repositories. Make sure these repositor
nd.
Computing distribution upgrade...

The following 126 packages are going to be upgraded:
  cups-filters2 feh gdk-pixbuf-loader-libheif gzip ImageMagick ImageMagick-config-7-SUSE klogd lastlog2 libb
  libharfbuzz-gobject0 libharfbuzz-icu0 libharfbuzz-subset0 libheif1 libheif-aom libheif-dav1d libheif-ffmpe
  libinput-udev libixion-0_20-0 liblastlog2-2 liblightdm-gobject-1-0 liblz1 libMagickCore-7_Q16HDRI10 libMag
  libp11-kit0 libpackagekit-glib2-18 libprotobuf-c1 libreoffice libreoffice-base libreoffice-calc libreoffic
  libreoffice-impress libreoffice-l10n-en libreoffice-mailmerge libreoffice-math libreoffice-pyuno libreoffi
  libsrtp2-1 libssh4 libssh-config libstorage-ng1 libstorage-ng-lang libstorage-ng-ruby libuuid1 libvte-2_91
  lightdm lightdm-bash-completion lightdm-gtk-greeter lightdm-gtk-greeter-branding-openSUSE lightdm-gtk-gree
  MozillaFirefox mozilla-nss mozilla-nss-certs MozillaThunderbird MozillaThunderbird-openpgp-librnp ncurses-u
  PackageKit-backend-zypp PackageKit-branding-openSUSE PackageKit-gstreamer-plugin PackageKit-gtk3-module pi
  snapper snapper-lang snapper-zypp-plugin terminfo terminfo-base terminfo-iterm terminfo-screen thin-provis
  util-linux-systemd wsdd
```

Update Mechanism

The difference between MicroOS and Tumbleweed is “architectural philosophy”.

To put it in an analogy:

MicroOS separates the act of updating from the act of living. Your running system is never contaminated by the update process. Tumbleweed intertwines them the system breathes and updates in the same lung.

- MicroOS: prepares an updated system state transactionally and activates it after reboot.
- Tumbleweed: performs package updates directly against the installed system using Zypper.

To put it simple, one values isolation; the latter values immediacy.

Rollback Capability (MicroOS)

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1

```
microos@localhost:~> sudo transactional-update pkg install nginx
tukit 6.1.3 started
Options: --log=console -c10 open
Using 'snapper' as snapshot manager.
Using snapshot 10 as base for new snapshot 11.
ID: 11
Transaction completed.
tukit 6.1.3 started
Options: --log=console callext 11 zypper -R {} install nginx
Using 'snapper' as snapshot manager.
Executing `zypper -R /tmp/transactional-update-EnwyoF install nginx`:
```

2

```
microos@localhost:~> sudo snapper list
  #  | Type   | Pre # | Date                            | User | Used Space | Cleanup | Description           | Userdata
-----+--------+-------+---------------------------------+------+------------+---------+-----------------------+---------
 0   | single |       |                                 | root |            |         | current               |
 1   | single |       | Mon 21 Sep 2026 08:16:42 AM UTC | root |    2.24 GiB|         | first root filesystem |
 2   | single |       | Sun 27 Sep 2026 10:25:06 AM UTC | root |    2.63 GiB| number  | Snapshot Update of #1 |
 3   | single |       | Sun 27 Sep 2026 10:49:29 AM UTC | root |    1.04 MiB| number  | Snapshot Update of #1 |
 8   | single |       | Sun 27 Sep 2026 12:01:56 PM UTC | root |    4.04 MiB| number  | Snapshot Update of #3 |
 9   | single |       | Sun 27 Sep 2026 12:02:33 PM UTC | root |    4.04 MiB| number  | Snapshot Update of #3 |
10-  | single |       | Sun 27 Sep 2026 12:02:58 PM UTC | root | 836.00 KiB| number  | Snapshot Update of #3 |
11+  | single |       | Sun 27 Sep 2026 12:12:15 PM UTC | root |    3.94 MiB| number  | Snapshot Update of #10|
microos@localhost:~>
```

3

```
microos@localhost:~> rpm -q nginx
nginx-1.31.6-1.1.x86_64
microos@localhost:~>
```

4

```
microos@localhost:~> sudo transactional-update rollback 2
tukit 6.1.3 started
Options: --log=console rollback 2
Using 'snapper' as snapshot manager.
Transaction completed.
microos@localhost:~>
```

5

```
microos@localhost:~> sudo snapper list
  #  | Type   | Pre # | Date                            | User | Used Space | Cleanup | Description           |
-----+--------+-------+---------------------------------+------+------------+---------+-----------------------+
 0   | single |       |                                 | root |            |         | current               |
 1   | single |       | Mon 21 Sep 2026 08:16:42 AM UTC | root |    2.24 GiB|         | first root filesystem |
 2+  | single |       | Sun 27 Sep 2026 10:25:06 AM UTC | root |    2.64 GiB|         | Snapshot Update of #1 |
 3   | single |       | Sun 27 Sep 2026 10:49:29 AM UTC | root |    1.04 MiB| number  | Snapshot Update of #1 |
 8   | single |       | Sun 27 Sep 2026 12:01:56 PM UTC | root |    4.04 MiB| number  | Snapshot Update of #3 |
 9   | single |       | Sun 27 Sep 2026 12:02:33 PM UTC | root |    4.04 MiB| number  | Snapshot Update of #3 |
10   | single |       | Sun 27 Sep 2026 12:02:58 PM UTC | root | 836.00 KiB| number  | Snapshot Update of #3 |
11   | single |       | Sun 27 Sep 2026 12:12:15 PM UTC | root |    3.19 MiB|         | Snapshot Update of #10|
12   | single |       | Sun 27 Sep 2026 12:18:42 PM UTC | root |    1.36 MiB| number  | Snapshot Update of #11|
13   | single |       | Sun 27 Sep 2026 12:23:33 PM UTC | root |    1.36 MiB| number  | Snapshot Update of #11|
microos@localhost:~> rpm -q nginx
package nginx is not installed
microos@localhost:~>
```

Rollback Capability (Tumbleweed)

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1

```
Home | openSUSE MicroOS | openSUSE Tumbleweed

File  Edit  View  Terminal  Tabs  Help
tumbleweed@localhost:~> rpm -q nginx
package nginx is not installed
tumbleweed@localhost:~> sudo zypper install nginx
Loading repository data...
Reading installed packages...
Resolving package dependencies...

The following 6 recommended packages were automaticall
  luajit luajit-cjson lua-resty-mysql lua-resty-redis

The following 10 NEW packages are going to be installe
  lua-interpreter luajit luajit-cjson lua-resty-core l

10 new packages to install.

Package download size:     1.6 MiB

Package install size change:
              |       4.9 MiB  required by packages th
    4.9 MiB   |   -       0 B    released by packages th

Backend:  classic_rpmtrans
Continue? [y/n/v/...? shows all options] (y): y
Preloading: lua-resty-lrucache-0.15-1.4.noarch.rpm [do
Preloading: lua-interpreter-5-3.3.noarch.rpm [done]
```

2

```
tumbleweed@localhost:~> rpm -q nginx
nginx-1.31.6-1.1.x86_64
tumbleweed@localhost:~> sudo snapper list
 #  | Type   | Pre # | Date                             | User
----+--------+-------+----------------------------------+------
0   | single |       |                                  | root
1*  | single |       | Mon 21 Sep 2026 09:02:55 AM WIB  | root
2   | single |       | Mon 21 Sep 2026 09:41:40 AM WIB  | root
3   | pre    |       | Sun 27 Sep 2026 05:24:35 PM WIB  | root
4   | post   |     3 | Sun 27 Sep 2026 05:24:41 PM WIB  | root
5   | pre    |       | Sun 27 Sep 2026 06:53:56 PM WIB  | root
6   | post   |     5 | Sun 27 Sep 2026 06:54:50 PM WIB  | root
7   | pre    |       | Sun 27 Sep 2026 07:43:52 PM WIB  | root
8   | post   |     7 | Sun 27 Sep 2026 07:43:55 PM WIB  | root
```

3

```
Home | openSUSE MicroOS | openSUSE Tumbleweed
                                        Terminal - tumbleweed@
File  Edit  View  Terminal  Tabs  Help
tumbleweed@localhost:~> rpm -q nginx
package nginx is not installed
tumbleweed@localhost:~> sudo snapper list
 #  | Type   | Pre # | Date                             | User | Used Space | Clea
----+--------+-------+----------------------------------+------+------------+-----
0   | single |       |                                  | root |            |
1   | single |       | Mon 21 Sep 2026 09:02:55 AM WIB  | root |  768.00 KiB | numb
2   | single |       | Mon 21 Sep 2026 09:41:40 AM WIB  | root |  110.61 MiB | numb
3   | pre    |       | Sun 27 Sep 2026 05:24:35 PM WIB  | root |  304.00 KiB | numb
4   | post   |     3 | Sun 27 Sep 2026 05:24:41 PM WIB  | root |    4.44 MiB | numb
5   | pre    |       | Sun 27 Sep 2026 06:53:56 PM WIB  | root |    1.54 MiB | numb
6   | post   |     5 | Sun 27 Sep 2026 06:54:50 PM WIB  | root |   73.95 MiB | numb
7   | pre    |       | Sun 27 Sep 2026 07:43:52 PM WIB  | root |   96.61 MiB | numb
8   | post   |     7 | Sun 27 Sep 2026 07:43:55 PM WIB  | root |  768.00 KiB | numb
9   | single |       | Sun 27 Sep 2026 07:46:16 PM WIB  | root |  752.00 KiB | numb
10  | single |       | Sun 27 Sep 2026 07:46:16 PM WIB  | root |   48.00 KiB | numb
11  | single |       | Sun 27 Sep 2026 07:49:23 PM WIB  | root |   32.00 KiB | numb
12* | single |       | Sun 27 Sep 2026 07:49:23 PM WIB  | root |  412.00 KiB |
tumbleweed@localhost:~> sorry i missed the "sudo snapper rollback 3" capture :D
```

Rollback Capability

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Both systems can provide rollback through Btrfs and Snapper.

Therefore, rollback itself should not be treated as the unique advantage of MicroOS. The research instead evaluates how rollback integrates with the update architecture.

- On MicroOS, rollback is woven into the transactional operating model in other words it is the default escape hatch.
- On Tumbleweed, rollback is a safety net you must weave yourself. The question is not whether you can go back; it is whether the system expects you to need to!

Persistence Resistance (MicroOS)

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1

```
Home | openSUSE MicroOS | openSUSE Tumbleweed
New Tab  Split View

microos@localhost:~> sudo mv research-persistence.service /etc/systemd/system

microos@localhost:~> cat /etc/systemd/system/research-persistence.service
[Unit]
Description=Research Persistence Test

[Service]
Type=oneshot
ExecStart=/usr/bin/touch /tmp/research-persistence-executed

[Install]
WantedBy=multi-user.target
microos@localhost:~>
```

2

```
Home | openSUSE MicroOS | openSUSE Tumbleweed
~ : bash — Konsole
New Tab  Split View  Copy  Paste  Fin

microos@localhost:~> sudo systemctl daemon-reload

microos@localhost:~> sudo systemctl enable research-persistence.service
Failed to enable unit: Unit research-persistence.service does not exist
microos@localhost:~> sudo systemctl enable research-persistence.service
Failed to enable unit: Unit research-persistence.service does not exist
microos@localhost:~> systemctl is-enabled research-persistence.service
not-found
microos@localhost:~> systemctl status research-persistence.service
Unit research-persistence.service could not be found.
microos@localhost:~>
```

Persistence Resistance (Tumbleweed)

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```
tumbleweed@localhost:~> sudo nano /etc/systemd/system/research-persistence.service

tumbleweed@localhost:~> cat /etc/systemd/system/research-persistence.service
[Unit]
Description=Research Persistence Test

[Service]
Type=oneshot
ExecStart=/usr/bin/touch /tmp/research-persistence-executed

[Install]
WantedBy=multi-user.target
tumbleweed@localhost:~>
```

1

```
tumbleweed@localhost:~> sudo systemctl enable research-persistence.service
Created symlink '/etc/systemd/system/multi-user.target.wants/research-persistence.service' → '/etc/systemd/syste
tumbleweed@localhost:~> systemctl is-enabled research-persistence.service
enabled
```

2

Terminal - tumb

File Edit View Terminal Tabs Help

```
tumbleweed@localhost:~> sudo systemctl start research-persistence.service

tumbleweed@localhost:~> sudo reboot
```

3

Terminal - tumbleweed

File Edit View Terminal Tabs Help

```
tumbleweed@localhost:~> ls -l /tmp/research-persistence-executed
-rw-r--r--. 1 root root 0 Sep 27 21:12 /tmp/research-persistence-executed
tumbleweed@localhost:~>
```

4

Persistence Resistance

The persistence resistance test does not really work on MicroOS, yet it works perfectly on Tumbleweed!

The objective is not to claim that MicroOS eliminates persistence.

Instead, we test whether an attacker can establish conventional system-level persistence through direct modification of the operating system.

- MicroOS raises the difficulty of modifying the immutable system state. The attacker cannot simply drop a file into /usr or expect /etc to behave like permanent clay.
- Tumbleweed provides conventional mechanisms for persistent system modification.

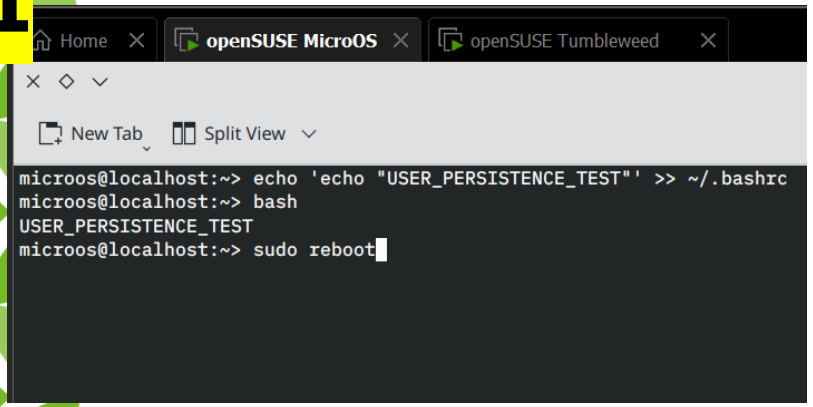
The immutable paradigm does not make persistence impossible but it forces the attacker to work harder, louder, and through different channels!

User-Space Persistence

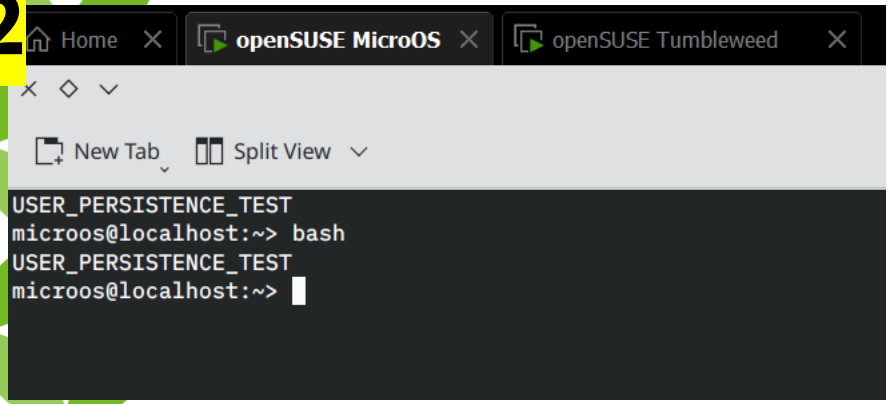
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MicroOS

1

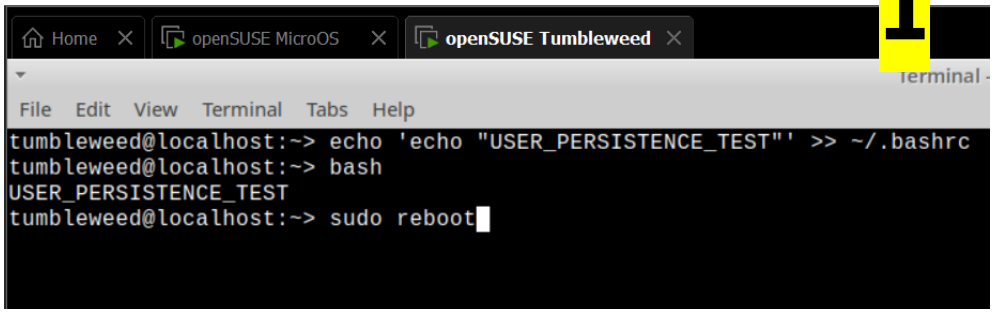


2

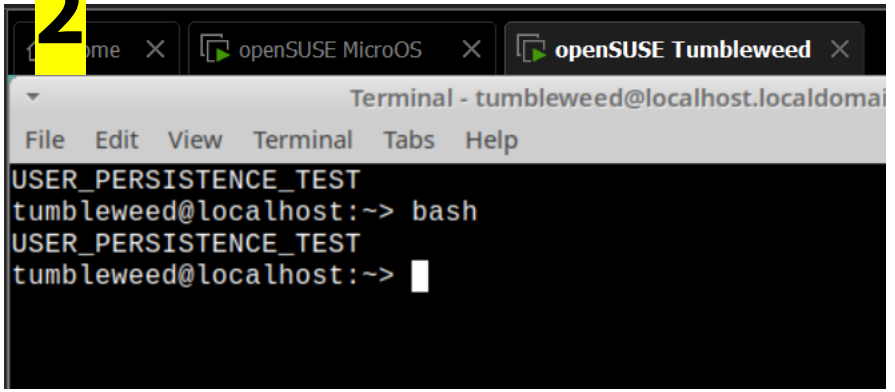


Tumbleweed

1



2



User-Space Persistence

The output is identical on both machines. The message prints. The persistence survives.

The user's home directory is a sovereign nation, and immutability has no embassy there.

This experiment demonstrates an important limitation of immutability. Protecting the operating-system image does not automatically protect user-space environments.

The castle walls may be impenetrable, but if the attacker is already inside the guest chambers, the walls do not matter. Immutable OS does not equal immutable user environment.

This is a critical insight from the cybersecurity angle: immutability shifts the battlefield, but it does not end “the war”.

System Drift (MicroOS)

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New Tab Split View

```
microos@localhost:~> rpm -qa | sort > ~/packages-baseline.txt
microos@localhost:~> sudo find /etc -type f -printf '%p\n' | sort > ~/etc-baseline.txt

microos@localhost:~>
microos@localhost:~> sudo find /etc -type f -exec sha256sum {} \; | sort > ~/etc-hashes-baseline.t
microos@localhost:~> sudo transactional-update pkg install nginx curl git
```

```
Transaction completed.
microos@localhost:~> sudo reboot
```

```
microos@localhost:~> rpm -qa | sort > ~/packages-current.txt
microos@localhost:~> sudo find /etc -type f -printf '%p\n' | sort > ~/etc-current.txt
microos@localhost:~> sudo find /etc -type f -exec sha256sum {} \; | sort > ~/etc-hashes-current.txt
microos@localhost:~> diff ~/packages-baseline.txt ~/packages-current.txt
123a124,125
> git-2.55.0-3.2.x86_64
> git-core-2.55.0-3.2.x86_64
559a562
> libgd3-2.3.3-6.1.x86_64
1041a1045
> libsha1detectcoll1-1.0.3-5.7.x86_64
1274a1279
> nginx-1.31.6-1.1.x86_64
1318a1324,1325
> perl-Error-0.170.300-1.5.noarch
> perl-Git-2.55.0-3.2.x86_64
microos@localhost:~> diff ~/etc-baseline.txt ~/etc-current.txt
17a18
> /etc/bash_completion.d/git-prompt
106a108
> /etc/logrotate.d/nginx
127a130,144
> /etc/nginx/fastcgi.conf
> /etc/nginx/fastcgi.conf.default
> /etc/nginx/fastcgi_params
> /etc/nginx/fastcgi_params.default
> /etc/nginx/koi-utf
> /etc/nginx/koi-win
> /etc/nginx/mime.types
> /etc/nginx/mime.types.default
> /etc/nginx/nginx.conf
> /etc/nginx/nginx.conf.default
> /etc/nginx/scgi_params
> /etc/nginx/scgi_params.default
> /etc/nginx/uwsgi_params
> /etc/nginx/uwsgi_params.default
> /etc/nginx/win-utf
161a179
> /etc/profile.d/git.csh
microos@localhost:~> diff ~/etc-hashes-baseline.txt ~/etc-hashes-current.txt
3c3
< 00c324323738379d32a242ef967d15020ff2825851a9a6671a646ad501c98ff1  /etc/group
---
> 00c324323738379d32a242ef967d15020ff2825851a9a6671a646ad501c98ff1  /etc/group-
10c10,11
< 01ad4422d36a4c12291eea1045aa11f035a0439278651ef5097d1559d291bbbf  /etc/shadow-
---
> 015cb581c2eb84b1a1ac9b575521d5881f791f632bfa62f34b26ba97d70c0d4f  /etc/nginx/uwsgi_params
> 015cb581c2eb84b1a1ac9b575521d5881f791f632bfa62f34b26ba97d70c0d4f  /etc/nginx/uwsgi_params.default
72a74
```

System Drift (Tumbleweed)

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Terminal - tumbleweed@localhost.localdomain:~

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```
tumbleweed@localhost:~> rpm -qa | sort > ~/packages-baseline.txt
tumbleweed@localhost:~> sudo find /etc -type f -printf '%p\n' | sort > ~/etc-bas
eline.txt
tumbleweed@localhost:~> sudo find /etc -type f -exec sha256sum {} \; | sort > ~/
etc-hashes-baseline.txt
tumbleweed@localhost:~> sudo zypper install nginx curl git
```

```
tumbleweed@localhost:~> sudo reboot
```

```
tumbleweed@localhost:~> rpm -qa | sort > ~/packages-current.txt
tumbleweed@localhost:~> sudo find /etc -type f -printf '%p\n' | sort > ~/etc-current.txt
tumbleweed@localhost:~> sudo find /etc -type f -exec sha256sum {} \; | sort > ~/etc-hashes-current.txt
tumbleweed@localhost:~> diff ~/packages-baseline.txt ~/packages-current.txt
179a180,182
> git-2.55.0-3.2.x86_64
> git-core-2.55.0-3.2.x86_64
> git-email-2.55.0-3.2.x86_64
1071a1075
> libsha1detectcoll1-1.0.3-5.7.x86_64
1473a1478
> perl-Authen-SASL-2.210.0-1.1.noarch
1478a1484,1485
> perl-Crypt-URandom-0.550.0-1.4.x86_64
> perl-Digest-HMAC-1.50.0-1.6.noarch
1481a1489
> perl-Error-0.170.300-1.5.noarch
1485a1494
> perl-Git-2.55.0-3.2.x86_64
1498a1508
> perl-MailTools-2.220.0-1.6.noarch
1501a1512
> perl-Net-SMTP-SSL-1.04-1.34.noarch
```

```
tumbleweed@localhost:~> diff ~/etc-baseline.txt ~/etc-current.txt
19a20
> /etc/bash_completion.d/git-prompt
246a248
> /etc/profile.d/git.csh
tumbleweed@localhost:~> diff ~/etc-hashes-baseline.txt ~/etc-hashes-current.txt
512a513
> 5fa8bf2b4233104c0c7820bde4899cfea00e60f6ab7a2c3fe7b0f341ecc0648e  /etc/ld.so.cache
990a992
> b29ed219f94a729f7122c351376ce335462292242b95aceb39d3a43312897100  /etc/bash_completion.d/git-prompt
1013d1014
< b48cac029304f47f820bb0b8a6da02d7fd8e8b043ccc072e0ade67f6ad31c473  /etc/ld.so.cache
1203a1205
> d4443f40f88de24b006b92d2771aedca8c4bfb7135b0a52cdb375adf0c50322d  /etc/profile.d/git.csh
tumbleweed@localhost:~>
```

System Drift

MicroOS's drift is surgical. You can look at the difference and understand precisely what changed, because the system only changes through deliberate, transactional acts.

MicroOS is designed to reduce uncontrolled drift by treating the operating system as a managed system state.

Tumbleweed's drift is organic: alive, growing, and increasingly unique to this specific machine.

Tumbleweed allows administrators to modify the system directly, providing flexibility but increasing the possibility of configuration divergence.

Administrative Usability

MicroOS provides stronger control over system state, but administrative tasks can require transactional updates and reboots. The administrator must learn a new rhythm plan, transact, reboot, verify.

Tumbleweed provides a more familiar Linux administration model, allowing administrators to modify packages and configurations directly, iterate rapidly, and troubleshoot without interruption.

The question for every organization is: what is more expensive? The time spent rebooting, or the time spent recovering from a broken mutable system at 3 AM?

Yeah, you can answer this by yourself … :D

Comparison Matrix

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Category	Main Command	MicroOS	Tumbleweed
Root Filesystem Immutablilty	findmnt + touch	/usr is read-only; root cannot touch system binaries	Fully writable; root has unrestricted access
Update-Mechanism	transactional-update	Transactional	Direct package update
Rollback Capability	snapper	transactional-update rollback — integrated, automatic	snapper rollback — manual, requires configuration
Persistence Resistance	systemctl	Restricted — direct system modification blocked	Full — conventional persistence mechanisms work
User-Space persistence	.bashrc	Still possible but more limited	Vulnerable
System Drift	diff	Low — snapshot baselines enforce predictability	High — incremental changes accumulate organically
Administrative Usability	Task timing	Steeper learning curve; predictable, reproducible state	Familiar; flexible; harder to reproduce exactly

Immutable OS != Secure OS



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Immutability should not be interpreted as a complete security boundary. It primarily changes the security properties of the operating-system layer.

An attacker may still compromise credentials, applications, user-space configuration, containers, or data.

Therefore, MicroOS should be evaluated as a resilience and integrity mechanism rather than as a complete defense against compromise.

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04 What concept all of this brings us to?

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Introducing “Secure By Design”

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Secure by Design is a proactive strategy that integrates cybersecurity into every stage of a product or service's creationfrom initial architecture to deployment and ongoing maintenance.



Introducing “Secure By Design”

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Instead of treating security as an afterthought, this model requires organizations to anticipate cyber threats early on and embed appropriate defenses directly into the design.

Its primary goal is to safeguard user privacy and data by building inherently resilient systems with fewer security flaws and maintaining those protections over the product's entire lifecycle.



Introducing “Secure By Design”

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- **Proactive Security Integration:** Cybersecurity is built into every stage of a product or service—from initial concept and architecture to deployment and ongoing maintenance.
- **Early Threat Modeling:** Organizations anticipate cyber threats right from the start, enabling them to design thoughtful, built-in defenses rather than patching issues later.
- **Holistic Organizational Shift:** Implementing this approach requires a company-wide commitment to security, rather than leaving it as an isolated IT task.
- **Vulnerability Reduction:** Products and services are engineered to contain fewer security flaws from day one, protecting consumer privacy and sensitive data.
- **Lifecycle Protection:** Security measures are actively maintained and updated throughout the entire lifespan of the product or service.





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